

MOTIVATION & RESEARCH QUESTION

WHY INDUSTRIAL EVENT-BASED DETECTION?

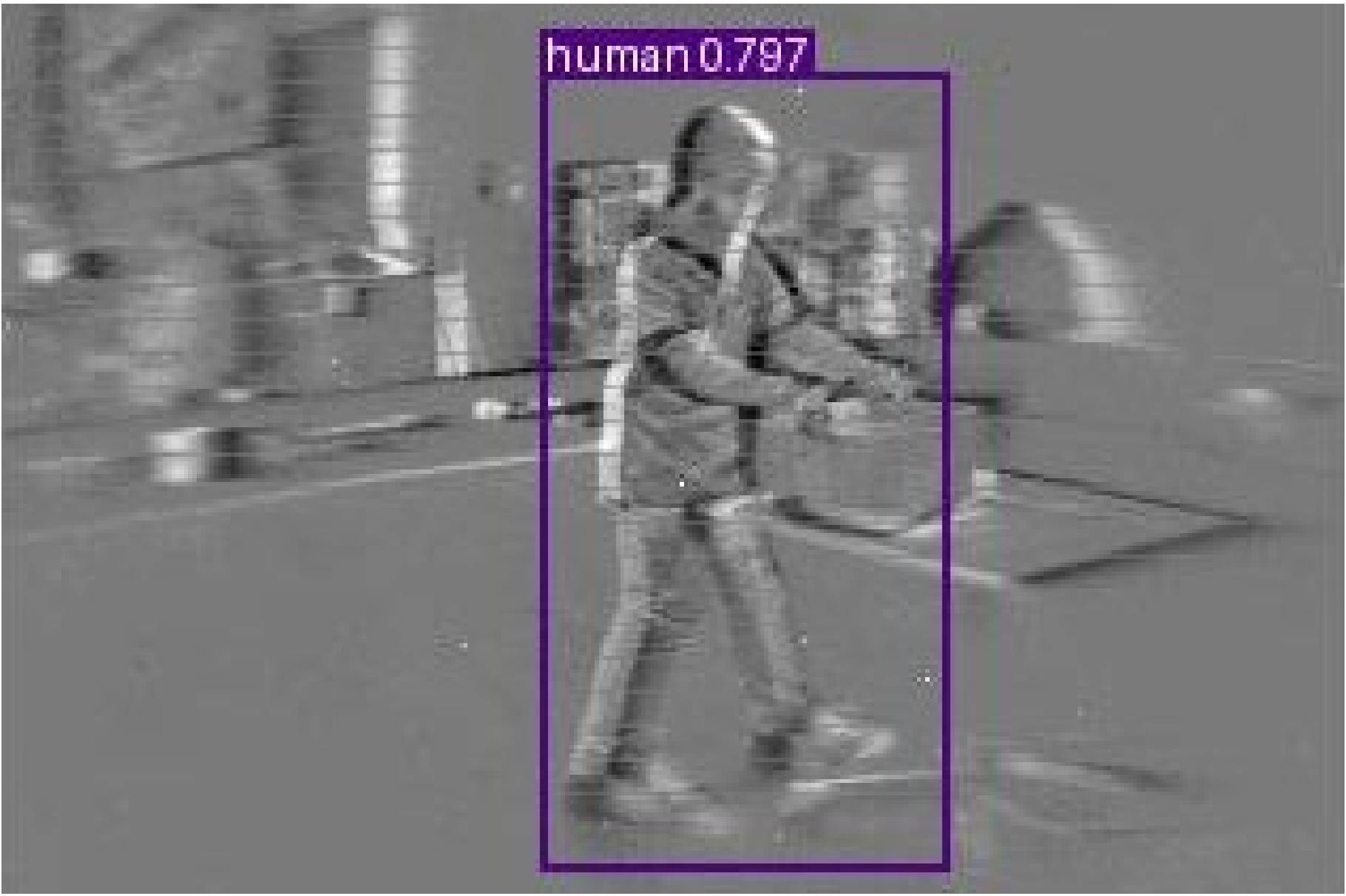
Event cameras offer high temporal resolution, high dynamic range, and reduced motion blur for fast robotic perception.

However, industrial multi-class recognition remains less explored than driving-focused event detection.

RESEARCH QUESTION

How much does recurrent temporal memory help, and how important is event-domain pretraining?

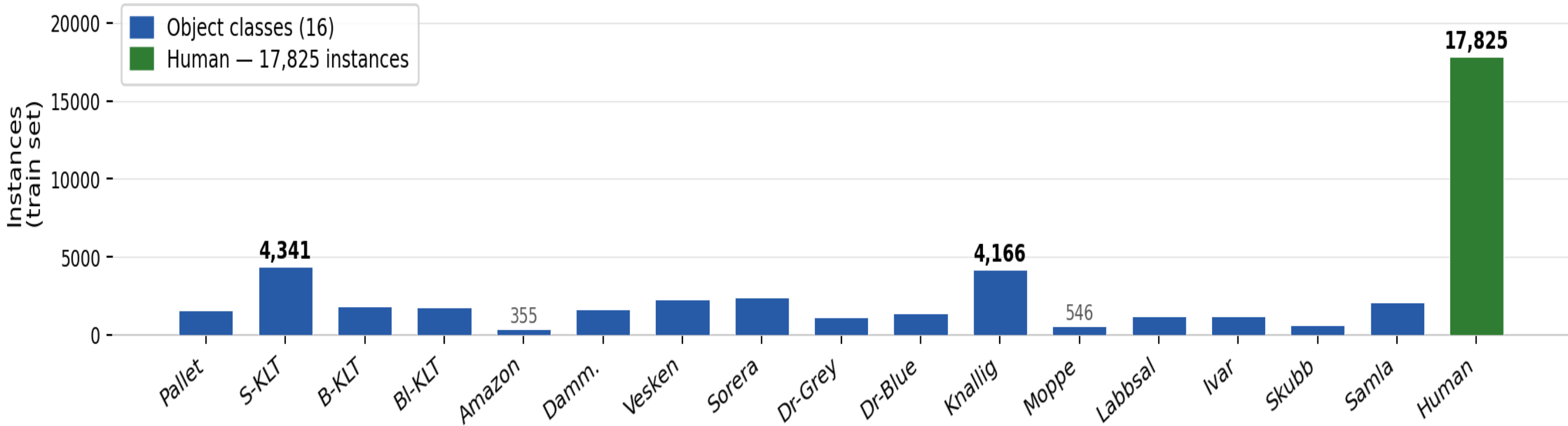
MTevent example



Industrial scenes include humans, objects, clutter, and class imbalance.

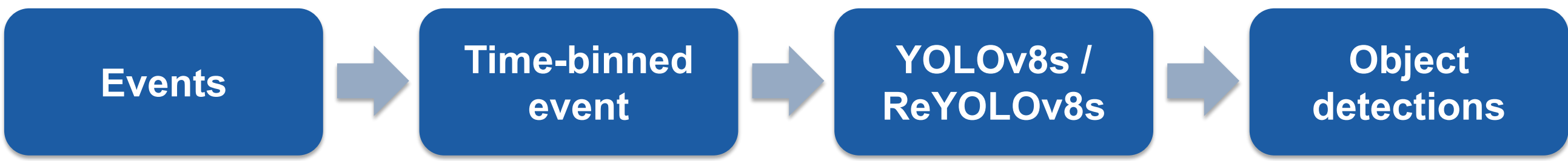
Dataset & contribution

- MTevent: industrial multi-class event dataset
- 17 object classes with strong imbalance
- Benchmark YOLOv8s against recurrent ReYOLOv8s



METHOD & EXPERIMENTAL SETUP

Detection pipeline



Events are accumulated into time-binned event frames, then evaluated with non-recurrent and recurrent detectors.

Model comparison

Baseline: YOLOv8s adapted for event-frame input; no recurrent temporal memory.

Recurrent model: ReYOLOv8s with ConvLSTM-based recurrent blocks and hidden states across event clips.

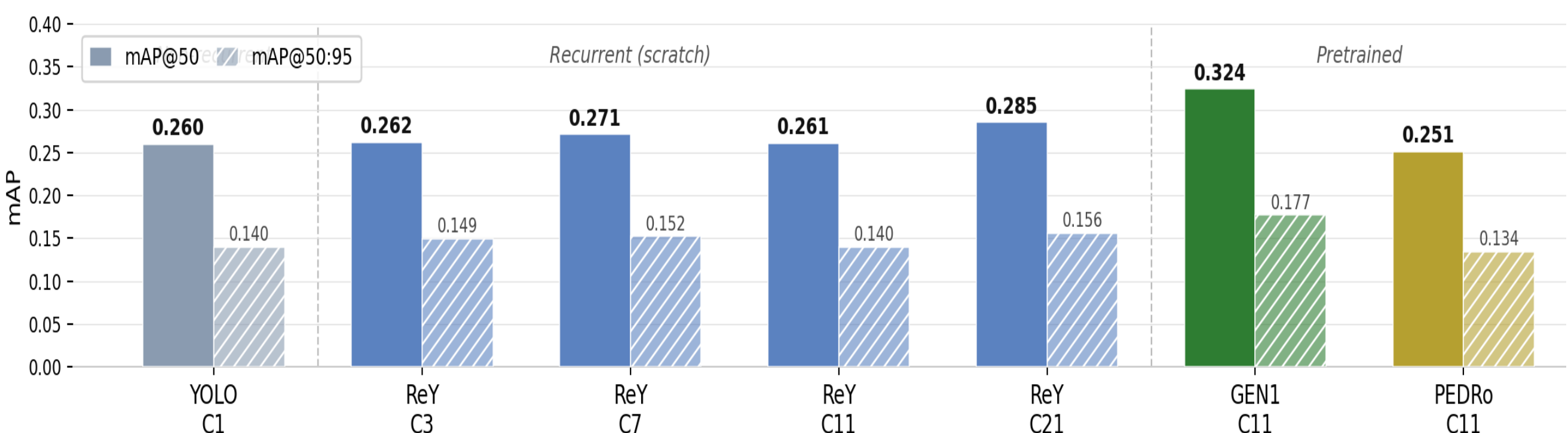
Experimental settings

Setting	Model	Initialization	Clip length
Baseline	YOLOv8s	Scratch	1
Recurrence	ReYOLOv8s	Scratch	3, 7, 11, 21
Pretraining	ReYOLOv8s	GEN1	3, 7, 11, 21
Pretraining	ReYOLOv8s	PEDRo	11, 21

Training/evaluation focus

- Clip length ablation: 3, 7, 11, 21
- Pretraining sources: GEN1 and PEDRo
- Metric: mAP50 on MTevent validation split

This design separates the effect of temporal memory from the effect of event-domain initialization.

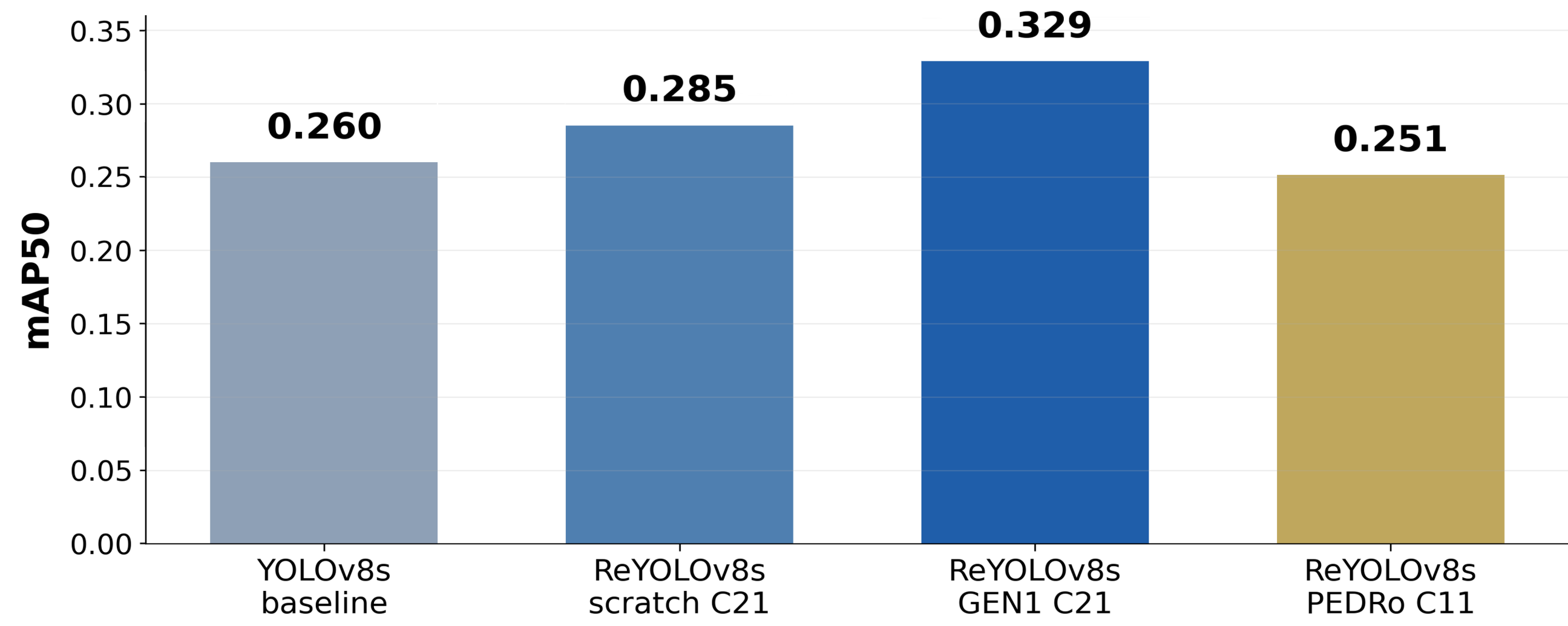


RESULTS & TAKEAWAYS

MAIN FINDING

Event-domain pretraining matters more than recurrence alone.

GEN1 pretraining gives the strongest improvement



Scratch recurrence: 0.260 → 0.285 mAP50. GEN1 fine-tuning reaches 0.329 mAP50.

Zero-shot qualitative examples



Zero-shot Human Detection



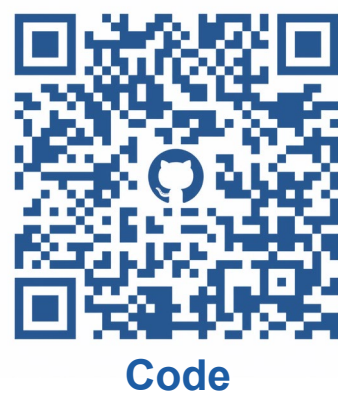
Zero-shot Detection Failure

Key takeaways

1. Recurrence gives a modest gain over the non-recurrent YOLOv8s baseline.
2. GEN1 event-domain pretraining gives the best performance.
3. Source-domain match is critical: PEDRo does not transfer well.
4. Remaining failures are linked to class imbalance and human-object interaction.



Paper



Code